NANO LETTERS

Letter

pubs.acs.org/NanoLett

Band Alignment in MoS_2/WS_2 Transition Metal Dichalcogenide Heterostructures Probed by Scanning Tunneling Microscopy and Spectroscopy

Heather M. Hill,[†,§,∥] Albert F. Rigosi,[†,§,∥] Kwang Taeg Rim,[‡] George W. Flynn,[‡] and Tony F. Heinz*[,†,§,∥]

[†]Departments of Physics and Electrical Engineering, Columbia University, New York, New York 10027, United States

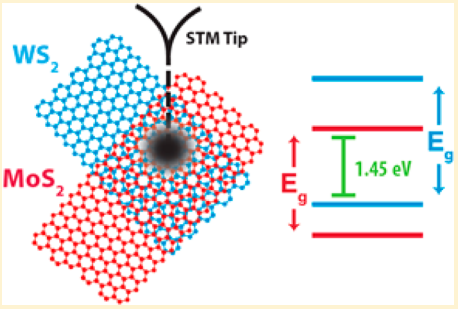
[‡]Department of Chemistry, Columbia University, New York, New York 10027, United States

[§]Department of Applied Physics, Stanford University, Stanford, California 94305, United States

[∥]SLAC National Accelerator Laboratory, 2575 Sand Hill Road, Menlo Park, California 94025, United States

Supporting Information

ABSTRACT: Using scanning tunneling microscopy (STM) and scanning tunneling spectroscopy (STS), we examine the electronic structure of transition metal dichalcogenide heterostructures (TMDCHs) composed of monolayers of MoS_2 and WS_2. STS data are obtained for heterostructures of varying stacking configuration as well as the individual monolayers. Analysis of the tunneling spectra includes the influence of finite sample temperature, yield information about the quasi-particle bandgaps, and the band alignment of MoS_2 and WS_2. We report the band gaps of MoS_2 (2.16 ± 0.04 eV) and WS_2 (2.38 ± 0.06 eV) in the materials as measured on the heterostructure regions and the general type II band alignment for the heterostructure, which shows an interfacial band gap of 1.45 ± 0.06 eV.



KEYWORDS: *2D materials, heterostructures, transition metal dichalcogenides, scanning tunneling microscopy, scanning tunneling spectroscopy*

The ability to combine layers into heterostructures exhibiting new electronic properties has contributed greatly both to fundamental studies and device applications. Two-dimensional (2D) van der Waals materials[1−4] offer new possibilities for the fabrication of heterostructures by combining individual monolayers of different materials. The interfaces of such heterostructure devices are inherently sharp on the atomic scale, and there is no need to match the lattices of the constituent layers since the materials do not have dangling bonds. In addition, one can use the angle of rotation between the sheets as a new degree of freedom.[5−9] The individual layers of 2D materials also offer a number of fascinating properties due to their reduced dimensionality. In this paper, we consider the family of semiconducting transition metal dichalcogenide (TMDC) layers as the building block for model heterostructures. Atomically thin layers in this family[10−13] display phenomena ranging from many-body effects,[14,15] coupling between carrier spin and valley degrees of freedom[16] to efficient light−matter interaction,[17,18] and notable excitonic behaviors.[19−21] Accordingly, TMDC heterostructures (TMDCHs) and their properties constitute a very active field research.[22−33]

A central question in the study of van-der-Waals TMDC heterostructures is how the electronic and optical properties of a heterostructure differ from that of the two individual layers. Recent attention has been particularly focused on the nature of the optically excited states of the heterostructure and the role charge transfer excitonic states.[22−26] These new states result from the relaxation of photoexcited electrons and holes into the adjacent layers. Such an excited state requires type II band alignment of the materials, as has been predicted theoretically.[34,35] The character of these charge transfer states, as well as the general problem of the charge transport properties across the heterostructure, requires knowledge of the band structure and band alignment of the heterostructure. Although optical signatures of the charge transfer state and general optical properties of heterostructures are currently being examined,[36] to date, the electronic structure of these TMDCHs has not been extensively explored.[37,38]

In this report we probe the electronic structure of the model TMDC heterostructure MoS_2/WS_2 with scanning tunneling microscopy (STM) and scanning tunneling spectroscopy (STS). STS has recently been used to examine the band gap and exciton binding energy of TMDC monolayers.[39−42] Here we find that, by analyzing the tunneling spectra for each of the monolayer constituents and their heterostructure regions, the band alignment of MoS_2 and WS_2 can be extracted. The band gaps of MoS_2 and WS_2, when they are measured as the top layer of their corresponding heterostructure, are 2.16 ± 0.04 eV

Received: March 8, 2016
Revised: May 25, 2016
Published: June 14, 2016

Downloaded via SHANGHAITECH UNIV on June 27, 2026 at 13:46:31 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

© 2016 American Chemical Society

DOI: 10.1021/acs.nanolett.6b01007
Nano Lett. 2016, 16, 4831−4837

and 2.38 ± 0.06 eV, respectively, and the type II band alignment for the heterostructure has an interfacial band gap of 1.45 ± 0.06 eV.

The TMDCHs investigated in this study were prepared on fused quartz substrates using mechanical exfoliation and transfer techniques.[36] Bulk TMDC crystals were exfoliated on fused quartz substrates and polypropylene carbonate transferable film. Monolayers were identified via photoluminescence (PL) and Raman spectroscopy, as well as atomic force microscopy (AFM). The TMDCHs were constructed by transferring one type of monolayer onto the other. In this fashion, we fabricated MoS_2/WS_2 heterostructures in both stacking orders, i.e., heterostructures with MoS_2 as the upper layer and others with WS_2.

For the STM/STS measurements, we also needed to provide electrical contact to the TMDCH. To avoid contamination of the TMDCH sample, we deposited a gold film (20 nm thickness) using evaporation through a shadow mask. A grid for transmission electron microscopy served as a convenient shadow mask. An image of a typical sample is shown in Figure 1; additional details about the technique are provided in the Supporting Information.

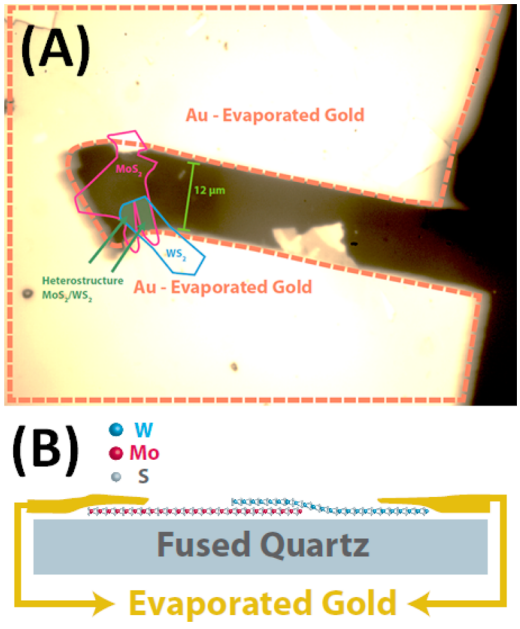


Figure 1. Optical image and schematic depiction of the samples used in the STM/STS measurements. (a) An optical image of the sample postgold deposition is shown. (b) An illustration to clarify how the gold is making contact with the heterostructure system.

Both STM and STS measurements were performed at room temperature on a commercial instrument (Omicron VT) with a tungsten tip.[43−45] The heterostructures were annealed in the STM chamber under ultrahigh vacuum (UHV) for 2 h at 450 K prior to the measurements. The STM images were recorded in constant current mode, with a tunneling voltage of $V = +1.5$ V. A representative STM image of an isolated monolayer of WS_2 presented in Figure 2 reveals the expected structure and periodicity.

The electronic structure of the monolayers and heterostructures was probed by STS measurements. These data were collected by turning off the feedback for the tip height and recording the tunneling current I while rapidly scanning the

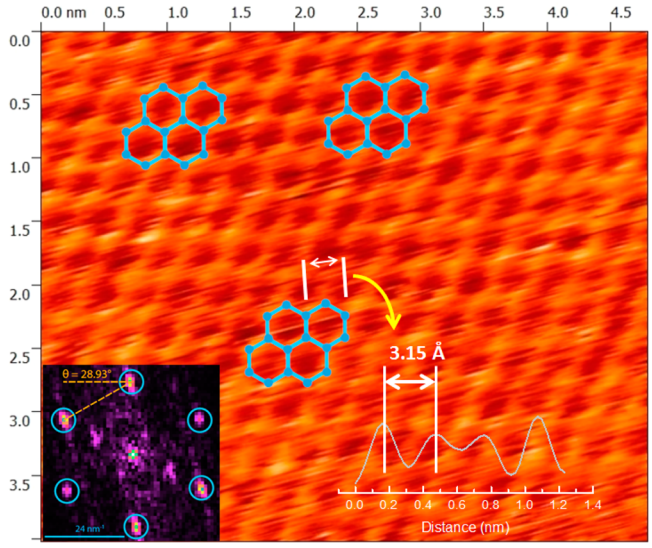


Figure 2. STM topography image for a WS_2 monolayer (5 nm × 4 nm area, $V = +1.5$ V, $I = 100$ pA). Representative honeycomb lattices are superimposed to guide the eye, and a representative topography image (lower right) verifies the expected lattice constant of WS_2. The Fourier transform of the real-space image (lower left) exhibits the expected periodicity.

tunneling voltage V. The feedback mechanism was then reengaged and the process was repeated at a different spatial location on the sample. The resulting $I-V$ spectra are numerically differentiated to obtain dI/dV spectra, which we present and discuss in this paper. Spectra are obtained for many lateral positions on surface and are averaged together to enhance the signal-to-noise ratio of the measurement. Overall, 80 sets of STS data were recorded for each sample. The structures were prepared so that measurements could be performed on each of the monolayer regions separately, as well as on the heterostructure region. TMDC monolayer STS data were taken to verify band gap values and, by extension, appropriate tunneling conditions with which we could probe heterostructure regions. To assess the reproducibility of the data, randomly selected subsets of data were averaged and compared in Figure 3. The key features in the data are seen to be consistent in the different measurements.

The spectra in Figure 3 exhibit the expected tunneling characteristics for semiconducting material, with a clear gap in the tunneling current for tip biases between the valence and conduction bands (VB and CB, respectively). Inspection of the spectra also reveals that the gap is significantly smaller for the heterostructures than for either of the separate monolayers, as would be expected for type II band alignment of a heterostructure.

We now introduce a model of the tunneling spectra that accounts for the electronic structure of the 2D layers, as well as the influence of the finite (300 K) measurement temperature. (Below we further consider the potential role of band bending in our experiment.) We model the tunneling current from a given band in the solid as[46]

$$I = \frac{4\pi e}{\hbar}\int_{-\infty}^{\infty}\left[\frac{1}{1+e^{\varepsilon - eV/kT}} - \frac{1}{1+e^{\varepsilon/kT}}\right] \times \rho_S(E_F - eV + \varepsilon)\rho_T(E_F + \varepsilon)T(z, V)d\varepsilon \quad (1)$$

DOI: 10.1021/acs.nanolett.6b01007
Nano Lett. 2016, 16, 4831−4837

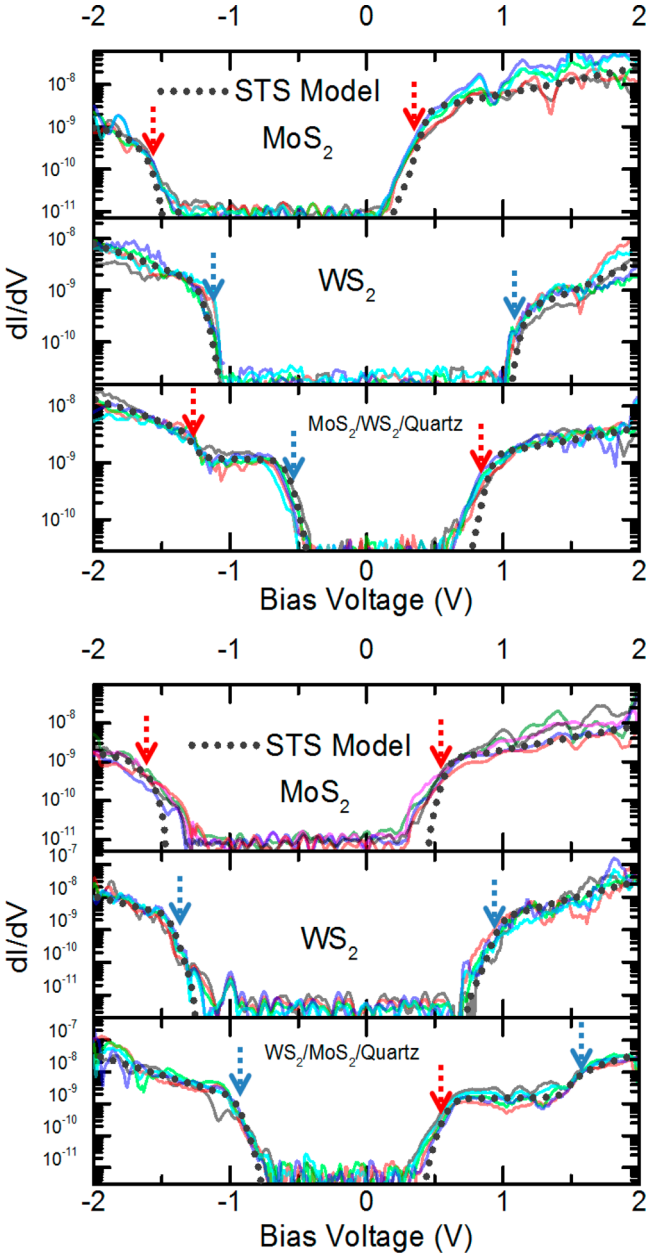


Figure 3. A set of STS data are shown for each of the three systems within each of the two vertical stacking orientations. The many colorful curves for each material are averages over 16 random measurements. The dotted curves are fits generated with a model, as described in the text, to extract values for the conduction and valence bands of monolayer MoS_2 and WS_2. The process is done for (top) the MoS_2/WS_2 and the (bottom) WS_2/MoS_2 stacking configurations. Red and blue dotted arrows are used to indicate the band onsets for MoS_2 and WS_2, respectively.

The tunneling current is seen to be an integral over all states in the tip at energy $E_F + \varepsilon$ (where E_F denotes the Fermi energy). In this expression, the contribution scales with the local density of states (LDOS) of the sample and tip, ρ_S and ρ_T, respectively, at the appropriate energies for an elastic tunneling process, as well as the tunneling probability T. The term in brackets describes the appropriate thermal occupation factors in the tip and sample. (See Supporting Information for further details.) We model the tunneling probability by an effective one-dimensional model:[47]

$$T \sim e^{-2z\sqrt{\frac{2m}{\hbar^2}\left(\frac{\Phi_s+\Phi_t}{2}-\frac{|eV|}{2}+E\right)}} \quad (2)$$

We apply the model above to describe tunneling into or out of a 2D band in the semiconductor. The LDOS is taken to be a step function, in accordance with the density of states of a 2D parabolic band; the LDOS for the metallic tip taken as constant. In evaluating the model, we made use of the following parameters: m (free electron mass), Φ_s = 5.1 eV (work function of monolayer MoS_2)[48] or 5.8 eV (work function of monolayer WS_2),[49] and Φ_t = 4.5 eV (work function of the tungsten metal tip). The decay constant of the tunneling probability depends on the parallel momentum of the electrons in the material, as reflected in the energy: $E = \frac{\hbar^2 k_{||}^2}{2m^*}$. This dependence would indicate that electrons from other parts of the band structure, such as the Γ point, would be more likely to tunnel into the sample than would electrons from the K point. The $k_{||}^2$ term can be found using the lattice constants, which yields $k_{||} \approx 1.33$ Å^{-1} at the K point. A further demonstration of this dependence can be seen in eq 2 by noticing how the lack of a parallel momentum term, representative of current flowing from the Γ point, can decrease the value of the term within the square root by about 30%. Our treatment of parallel momentum in the tunneling probability term is also implemented in another recent work.[47] The effective mass of the electron (m^*) are given the values of $0.35m_0$ and $0.3m_0$ for MoS_2 and WS_2, respectively.[50] The initial distance from the tip to the sample was approximated to be 5 Å, for reasons discussed below.

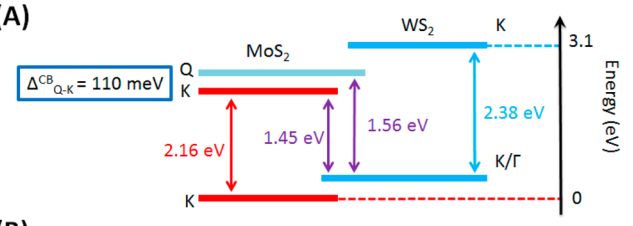
The analysis above describes the behavior for tunneling into or out of a single TMDC layer. To account for the additional tunneling processes in the heterostructure, additional terms were added to represent the second layer located 6 Å further below the tip. The final model for the heterostructure treated the current as tunneling into or out of either the Q or Γ point of the Brillouin zone in the bottom layer. We will discuss the origin of this choice for tunneling channels below.

The model is considered complete when a fitted model of the simulated current derivative successfully produces a curve, which when compared to the data, has a minimized least-squares regression. More information about this optimization process is provided in the Supporting Information. Models are constructed for each of the monolayer TMDCs in addition to the TMDCH. This makes for a total count of six models: three for the MoS_2/WS_2/quartz system and three for the WS_2/MoS_2/quartz system, all of which are presented in Figure 3. From the optimized models, we extract the values of the band gap for each of the materials shown in Figure 4. The MoS_2 and WS_2 quasi-particle band gaps, as measured on the heterostructure regions and without correction from tip-induced band bending, are found to be 2.19 ± 0.04 eV and 2.41 ± 0.06 eV, respectively.

In order to evaluate the effect of tip-induced band bending (TIBB), the computer program SEMITIP Version 6, introduced by R. Feenstra in 2011, along with the Film2 package, was utilized.[51−53] This program calculates the TIBB in a 2D film and includes a variety of parameters that can be approximately very well. For a list of all the relevant parameters used; see the Supporting Information. The charge density we used is higher than the typical exfoliated sample because samples that have undergone annealing are subject to additional doping of around 2×10^{12} cm^{-2}.[54] The calculated TIBB effect, in total, for each monolayer material, is about 30 meV, and since TIBB artificially enlarges the measured band gap, this

DOI: 10.1021/acs.nanolett.6b01007
Nano Lett. 2016, 16, 4831−4837

(A)



(B)

Material/ Parameter	Monolayer E_{g1} Band Gap	Monolayer E_{g2} Band Gap	Φ	m*	$k_{\|}$ (Å^{-1})
MoS_2	2.18 ± 0.05 eV	2.17 ± 0.04 eV	5.1 eV	0.35	1.32
WS_2	2.40 ± 0.06 eV	2.39 ± 0.06 eV	5.8 eV	0.3	1.33

Figure 4. Experimental band alignment in the heterostructures is schematically depicted with band gap values measured on the heterostructure region in panel a. A table of various important values for MoS_2 and WS_2 is presented in panel b, such as the measured monolayer band gaps in the two different stacking orientations, the work function, effective mass, and K-point parallel momentum. The E_{g1} and E_{g2} notations represent the systems where MoS_2 and WS_2 is the top layer, respectively. See text for further details.

value must be subtracted from the measurement. TIBB calculations done on the heterostructure region for the low-current onsets are very roughly approximated to be negligible (see Supporting Information for more details on numerical calculation and effects from varying charge carrier densities). All of the presented numbers henceforth have this value subtracted.

The assignment of the onsets in the STS data requires some delicate interpretation. If we are to consider the K points as too sensitive to measure, as in some instances in STS,[47] then we must consider other parts of the Brillouin zone. In the case of the CB of MoS_2, the Q point (between K and Γ) would have the first measurable onset energy,[30,34,35] meaning that the energy difference between the Q point in the CB and K point in the valence band would be 2.18 ± 0.05 eV, which is the measured gap on monolayer MoS_2 on quartz substrate. However, this value would not be consistent with previously measured values of the exciton binding energy.[19,21] The Q point is on the order of 250 meV higher in energy than the CB K point for monolayer MoS_2 on graphite,[47] and around 200 meV for isolated MoS_2,[14] and these values would require the K−K energy gap to be less than 2 eV. This K−K gap, when coupled to the 1.86 eV value of the exciton transition in monolayer MoS_2, would yield an incompatible binding energy of around 100 meV. Therefore, we interpret the two main onsets in MoS_2 as the K points in the Brillouin zone, verifying that our scanning parameters are sufficient to probe these points. Our interpretation is also consistent with previously reported STS results.[55]

In monolayer WS_2, the Γ point in the valence band is close in energy to the K point.[30,34,35] The CB can be interpreted as the K point, but in the valence band, the Γ point must also be considered as a potential significant contributor to the current. However, the onset of the Γ point is not entirely obvious in the data, but it is possible that the two points of the Brillouin zone are close enough such that the resolution at room temperature for both onsets is lost. Thus, we incorporate the Γ point in the model for the valence band by setting $k_{\|} = 0$. The band gap for WS_2 on quartz, 2.40 ± 0.06 eV, is compatible with what has been reported in the relevant literature.[21] Our capacity to measure the band gaps of the individual layers, and to use such measurement capabilities for the heterostructures, is now verified to within experimental uncertainty.

Upon inspection of the first heterostructure system, MoS_2 on WS_2 on quartz (top in Figure 3), the appearance of a third onset reduces the apparent band gap in the data, indicating that the band gaps align in a type II configuration, as predicted in some works.[30,37] Two of the onsets, the lone CB onset and the lower energy onset in the valence band, occur with a spacing of 2.16 ± 0.04 eV, which is interpreted as the CB and VB of the top layer of this heterostructure system—MoS_2. The additional onset in the valence band region can be attributed to the WS_2 layer beneath. Due to the high parallel momentum at the K point, the likelihood that we are probing the K point of WS_2 at an additional 6 Å further from the tip is quite low. However, the Γ point is a more-likely accessible feature of this system, and it is predicted that the Γ and K points in the valence band of WS_2 are nearly degenerate when the monolayer WS_2 is formed into a heterostructure with MoS_2.[22] If we assign this onset to be the Γ point of WS_2, then we measure an energy difference of 1.45 ± 0.06 eV from the valence band of WS_2 to the CB (K point) of MoS_2. To within the experimental uncertainty, this value also applies to the K−K energy gap and is in good agreement with theoretical predictions.[22,35] We expect a type II alignment where WS_2 has a higher energy CB and VB than MoS_2 because of their respective STS measurements. This band alignment is shown in Figure 4.

We can now turn our attention to the second configuration of TMDCH stacking, WS_2 on MoS_2, shown in Figure 3 (bottom). In addition to the top layer onsets, measured to be 2.38 ± 0.06 eV—in close correspondence with the value of the monolayer WS_2 band gap, a third onset appears lower in energy in the CB of the heterostructure system. This onset is a result of the MoS_2 layer beneath, and its assignment is slightly more complicated. The measured energy difference between this low-energy CB onset and the K/Γ point of WS_2 (since we are measuring on WS_2 in a heterostructure, these two VB onsets are still virtually degenerate) is 1.56 ± 0.06 eV. With the transitivity of the K−K energy gap in TMDCHs, as demonstrated by ref 22, this CB onset must not arise from the K point of MoS_2. As previously stated, the ability to probe a point in the band with high in-plane momentum is not very likely given the distance away from the tip. As shown in ref 47, the Q point is a stronger contributor of current than the K point. In monolayer MoS_2, the energy spacing between the Q−K points is close to 250 meV. This spacing is closer to 200 meV when the monolayer is not on graphite.[14] However, in this heterostructure system, band gap calculations indicate that this energy spacing will take a much lower value, but not degenerate with the K points.[22] Therefore, we assign this feature to the Q point of MoS_2 in the CB and find a Q−K energy spacing of approximately 110 meV. The reduction in the energy of Q point is predicted when probing the electronic structure of this TMDCH system.[22] Furthermore, the spacing between the CB Q and K point in bilayer MoS_2 is expected to be approximately 100 meV.[8] This Q−K energy spacing and overall band alignment may suggest that the Q-K spacing in MoS_2, as part of a heterostructure, is not as sensitive to crystallographic orientation as other parts of the Brillouin zone, such as the K or Γ point.

In regards to the band onsets in the TMDCHs, the overall lateral shifting of the heterostructure STS spectrum along the voltage axis can be partially attributed to an increase in the dielectric screening environment for both layers in both vertical

DOI: 10.1021/acs.nanolett.6b01007
Nano Lett. 2016, 16, 4831−4837

orientations. Another example of this shifting is presented in the Supporting Information of ref 38. Although the MoS_2/WS_2/quartz system shifts in one direction, the bottom layer in the WS_2/MoS_2/quartz system does not. For this case, electrostatic processes and charge transfer from WS_2 to MoS_2 may counter the effects of this shifting for the Q point in MoS_2. More investigation into the effects of charge transfer on the electronic band structure at 2D/2D interfaces is called for.

We can favorably compare our bandgap values of 2.16 ± 0.04 eV and 2.38 ± 0.06 eV for MoS_2 and WS_2, respectively, to recently reported STS results[22,38,55,56] and another system with type II band alignment.[57] Additionally, the binding energies we extract are within 50−100 meV of binding energies found in optical measurements.[19,21,58,59]

In summary, we used scanning tunneling microscopy (STM) and scanning tunneling spectroscopy (STS) to examine the electronic structure of transition metal dichalcogenide heterostructures (TMDCHs) composed of monolayers of MoS_2 and WS_2 in both possible vertical stacking configurations. The STS data we report gives insight on the band alignment and offset of MoS_2 and WS_2. The quasiparticle band gaps of MoS_2 (2.16 ± 0.04 eV) and WS_2 (2.38 ± 0.06 eV), as measured on a heterostructure system, were extracted with the optimization of an STS model using first principles. We also measure a heterostructure bandgap of 1.45 ± 0.06 eV between the CB K point of MoS_2 and VB K point of WS_2. Additionally, we are able to assign the energy difference between the Q and K points of the CB of MoS_2 as 110 meV.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.nanolett.6b01007.

Sample preparation, photoluminescence, Raman spectra, AFM images, and other measurements (PDF)

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: t.heinz@stanford.edu. Fax: 212-854-1909.

Author Contributions

H.M.H. and A.F.R. contributed equally to this work.

Funding

Support was provided by the National Science Foundation through grant DMR-1123894. H.M.H. and A.F.R. acknowledge funding from the National Science Foundation through the Integrated Graduate Education and Research Training Fellowship (DGE-1069240) and the Graduate Research Fellowship Program (DGE-1144155), respectively.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We would like to thank Tingyi Gu, Randall Feenstra, Xiaodong Zhu, and Abhay Pasupathy for fruitful discussions.

■ REFERENCES

(1) Novoselov, K. S.; Jiang, D.; Schedin, F.; Booth, T. J.; Khotkevich, V. V.; Morozov, S. V.; Geim, A. K. *Proc. Natl. Acad. Sci. U. S. A.* **2005**, *102*, 10451.

(2) Wang, Q.; Kalantar-Zadeh, K.; Kis, A.; Coleman, J. N.; Strano, M. S. Electronics and optoelectronics of two-dimensional transition metal dichalcogenides. *Nat. Nanotechnol.* **2012**, *7*, 699−712.

(3) Grigorieva, I. V.; Geim, A. K. van der Waals Heterostructures. *Nature* **2013**, *499*, 419−425.

(4) Butler, S. Z.; Hollen, S. M.; Cao, L.; Cui, Y.; Gupta, J. A.; Gutiérrez, H. R.; Heinz, T. F.; Hong, S. S.; Huang, J.; Ismach, A. F.; Johnston-Halperin, E.; Kuno, M.; Plashnitsa, V. V.; Robinson, R. D.; Ruoff, R. S.; Salahuddin, S.; Shan, J.; Shi, L.; Spencer, M. G.; Terrones, M.; Windl, W.; Goldberger, J. E. Progress, challenges, and opportunities in two-dimensional materials beyond graphene. *ACS Nano* **2013**, *7*, 2898−2926.

(5) Jariwala, D.; Sangwan, V. K.; Lauhon, L. J.; Marks, T. J.; Hersam, M. C. Emerging Device Applications for Semiconducting Two-Dimensional Transition Metal Dichalcogenides. *ACS Nano* **2014**, *8*, 1102−1120.

(6) Hunt, B.; Sanchez-Yamagishi, J. D.; Young, A. F.; Yankowitz, M.; LeRoy, B. J.; Watanabe, K.; Taniguchi, T.; Moon, P.; Koshino, M.; Jarillo-Herrero, P.; Ashoori, R. C. Massive Dirac fermions and Hofstadter butterfly in a van der Waals heterostructure. *Science* **2013**, *340*, 1427−1430.

(7) Dean, C. R.; Wang, L.; Maher, P.; Forsythe, C.; Ghahari, F.; Gao, Y.; Katoch, J.; Ishigami, M.; Moon, P.; Koshino, M.; Taniguchi, T.; Watanabe, K.; Shepard, K. L.; Hone, J.; Kim, P. Hofstadter's butterfly and the fractal quantum Hall effect in moiré superlattices. *Nature* **2013**, *497*, 598−602.

(8) van der Zande, A. M.; Kunstmann, J.; Chernikov, A.; Chenet, D. A.; You, Y.; Zhang, X.; Huang, P. Y.; Berkelbach, T. C.; Wang, L.; Zhang, F.; Hybertsen, M. S.; Muller, D. A.; Reichman, D. R.; Heinz, T. F.; Hone, J. C. Tailoring the electronic structure in bilayer molybdenum disulfide via interlayer twist. *Nano Lett.* **2014**, *14*, 3869−3875.

(9) Liu, K.; Zhang, L.; Cao, T.; Jin, C.; Qiu, D.; Zhou, Q.; Zettl, A.; Yang, P.; Louie, S. G.; Wang, F. Evolution of Interlayer Coupling in Twisted MoS_2 Bilayers. *Nat. Commun.* **2014**, *5*, 4966.

(10) Wilson, J. A.; Yoffe, A. D. The Transition Metal Dichalcogenides Discussion and Interpretation of the Observed Optical, Electrical and Structural Properties. *Adv. Phys.* **1969**, *18*, 193−335.

(11) Neville, R. A.; Evans, B. L. The Band Edge Excitons in 2H-MoS2. *Phys. Status Solidi B* **1976**, *73*, 597.

(12) Splendiani, A.; Sun, L.; Zhang, Y. B.; Li, T. S.; Kim, J.; Chim, C. Y.; Galli, G.; Wang, F. Emerging Photoluminescence in Monolayer MoS_2. *Nano Lett.* **2010**, *10*, 1271−1275.

(13) Mak, K. F.; Lee, C.; Hone, J.; Shan, J.; Heinz, T. F. Atomically Thin MoS_2: A New Direct-Gap Semiconductor. *Phys. Rev. Lett.* **2010**, *105*, 136805.

(14) Qiu, D. Y.; da Jornada, F. H.; Louie, S. G. Optical spectrum of MoS 2: Many-body effects and diversity of exciton states. *Phys. Rev. Lett.* **2013**, *111*, 216805.

(15) Spataru, C. D.; Ismail-Beigi, S.; Capaz, R. B.; Louie, S. G. Quasiparticle and excitonic effects in the optical response of nanotubes and nanoribbons Top. *Top. Appl. Phys.* **2007**, *111*, 195−227.

(16) Wang, G.; Bouet, L.; Lagarde, D.; Vidal, M.; Balocchi, A.; Amand, T.; Marie, X.; Urbaszek, B. Valley dynamics probed through charged and neutral exciton emission in monolayer WSe2. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2014**, *90*, 075413.

(17) Li, Y.; Chernikov, A.; Zhang, X.; Rigosi, A.; Hill, H.; Van Der Zande, A. M.; Chenet, D. A.; Shih, E.; Hone, J.; Heinz, T. F. Measurement of the optical dielectric function of monolayer transition-metal dichalcogenides: MoS2, MoSe2, WS2, and WSe2. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2014**, *90*, 205422.

(18) Zhao, W.; Ghorannevis, Z.; Chu, L.; Toh, M.; Kloc, C.; Tan, P.-H.; Eda, G. Evolution of Electronic Structure in Atomically Thin Sheets of WS_2 and WSe_2. *ACS Nano* **2013**, *7*, 791−797.

(19) Hill, H. M.; Rigosi, A. F.; Roquelet, C.; Chernikov, A.; Berkelbach, T. C.; Reichman, D. R.; Hybertsen, M. S.; Brus, L. E.; Heinz, T. F. Observation of excitonic Rydberg states in monolayer MoS2 and WS2 by photoluminescence excitation spectroscopy. *Nano Lett.* **2015**, *15*, 2992.

DOI: 10.1021/acs.nanolett.6b01007
Nano Lett. 2016, 16, 4831−4837

(20) Liu, G.-B.; Shan, W.-Y.; Yao, Y.; Yao, W.; Xiao, D. Three-Band Tight-Binding Model for Monolayers of Group-VIB Transition Metal Dichalcogenides. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2013**, *88*, 085433.
(21) Chernikov, A.; Berkelbach, T. C.; Hill, H. M.; Rigosi, A.; Li, Y.; Aslan, O. B.; Reichman, D. R.; Hybertsen, M. S.; Heinz, T. F. Exciton Binding Energy and Nonhydrogenic Rydberg Series in Monolayer WS_2. *Phys. Rev. Lett.* **2014**, *113*, 076802.
(22) Chiu, M.-H.; Zhang, C.; Shiu, H. W.; Chuu, C.-P.; Chen, C.-H.; Chang, C.-Y. S.; Chen, C.-H.; Chou, M.-Y.; et al. Determination of band alignment in the single-layer MoS_2/WSe_2 heterojunction. *Nat. Commun.* **2015**, *6*, 7666.
(23) Yu, Y.; Hu, S.; Su, L.; Huang, L.; Liu, Y.; Jin, Z.; Purezky, A. A.; Geohegan, D. B.; Kim, K. W.; Zhang, Y.; et al. Equally Efficient Interlayer Exciton Relaxation and Improved Absorption in Epitaxial and Non-epitaxial MoS_2/WS_2 Heterostructures. *Nano Lett.* **2015**, *15*, 486.
(24) Hong, X.; Kim, J.; Shi, S. F.; Zhang, Y.; Jin, C.; Sun, Y.; Tongay, S.; Wu, J.; Zhang, Y.; Wang, F. Ultrafast Charge Transfer in Atomically Thin MoS2/WS2 Heterostructures. *Nat. Nanotechnol.* **2014**, *9*, 682−686.
(25) Rivera, P.; Schaibley, J. R.; Jones, A. M.; Ross, J. S.; Wu, S.; Aivazian, G.; Klement, P.; Ghimire, N. J.; Yan, J.; Mandrus, D. G. Observation of Long-Lived Interlayer Excitons in Monolayer $MoSe_2$-WSe_2 Heterostructures. *Nat. Commun.* **2015**, *6*, 6242.
(26) Fang, H.; Battaglia, C.; Carraro, C.; Nemsak, S.; Ozdol, B.; Kang, J. S.; Bechtel, H. A.; Desai, S. B.; Kronast, F.; Unal, A. A.; et al. Strong Interlayer Coupling in van der Waals Heterostructures Built from Single-Layer Chalcogenides. *Proc. Natl. Acad. Sci. U. S. A.* **2014**, *111*, 6198.
(27) Tongay, S.; Fan, W.; Kang, J.; Park, J.; Koldemir, U.; Suh, J.; Narang, D. S.; Liu, K.; Ji, J.; Li, J.; Sinclair, R.; Wu, J. Tuning Interlayer Coupling in Large-Area Heterostructures with CVD-Grown MoS2 and WS2Monolayers. *Nano Lett.* **2014**, *14*, 3185−3190.
(28) Ceballos, F.; Bellus, M. Z.; Chiu, H.; Zhao, H. Ultrafast Charge Separation and Indirect Exciton Formation in a MoS2−MoSe2 van der Waals Heterostructure. *ACS Nano* **2014**, *8*, 12717.
(29) Shim, G. W.; Yoo, K.; Seo, S. B.; Shin, J.; Jung, D. Y.; Kang, I. S.; Ahn, C. W.; Cho, B. J.; Choi, S. Y. Large-Area Single-Layer MoSe2 and Its van der Waals Heterostructures. *ACS Nano* **2014**, *8*, 6655−6662.
(30) Terrones, H.; Lopez-Urias, F.; Terrones, M. Novel hetero-layered materials with tunable direct band gaps by sandwiching different metal disulfides and diselenides. *Sci. Rep.* **2013**, *3*, 1549.
(31) Chiu, M.-H.; Li, M.-Y.; Zhang, W.; Hsu, W.-T.; Chang, W.-H.; Terrones, M.; Terrones, H.; Li, L.-J. Spectroscopic Signatures for Interlayer Coupling in MoS2 WSe2 van der Waals Stacking. *ACS Nano* **2014**, *8*, 9649−9656.
(32) Lin, Y. C.; Chang, C. Y. S.; Ghosh, R. K.; Li, J.; Zhu, H.; Addou, R.; Diaconescu, B.; Ohta, T.; Peng, X.; Lu, N.; Kim, M.; Robinson, J. T.; Wallace, R. M.; Mayer, T. S.; Datta, S.; Li, L.-J.; Robinson, J. A. Atomically Thin Heterostructures Based on Single-Layer Tungsten Diselenide and Graphene. *Nano Lett.* **2014**, *14*, 6936−6941.
(33) Huo, N.; Kang, J.; Wei, Z.; Li, S. S.; Li, J.; Wei, S. H. Novel and Enhanced Optoelectronic Performances of Multilayer MoS2−WS2 Heterostructure Transistors. *Adv. Funct. Mater.* **2014**, *24*, 7025−7031.
(34) Rigosi, A. F.; Hill, H. M.; Li, Y.; Chernikov, A.; Heinz, T. F. Probing Interlayer Interactions in Transition Metal Dichalcogenide Heterostructures by Optical Spectroscopy: MoS2/WS2 and MoSe2/WSe2. *Nano Lett.* **2015**, *15*, 5033.
(35) Kang, J.; Tongay, S.; Zhou, J.; Li, J.; Wu, J. Band offsets and Heterostructures of Two-Dimensional Semiconductors. *Appl. Phys. Lett.* **2013**, *102*, 012111.
(36) McDonnell, S.; Azcatl, A.; Addou, R.; Gong, C.; Battaglia, C.; Chuang, S.; Cho, K.; Javey, A.; Wallace, R. M. Hole Contacts on Transition Metal Dichalcogenides: Interface Chemistry and Band Alignments. *ACS Nano* **2014**, *8*, 6265−6272.
(37) Chiappe, D.; Scalise, E.; Cinquanta, E.; Grazianetti, C.; van den Broek, B.; Fanciulli, M.; Houssa, M.; Molle, A. Two-Dimensional Si Nanosheets with Local Hexagonal Structure on a MoS_2 Surface. *Adv. Mater.* **2014**, *26*, 2096−2101.
(38) Ugeda, M. M.; Bradley, A. J.; Shi, S.-F.; da Jornada, F. H.; Zhang, Y.; Qiu, D. Y.; Mo, S.-K.; Hussain, Z.; Shen, Z.-X.; Wang, F.; et al. Giant Bandgap Renormalization and Excitonic Effects in a Monolayer Transition Metal Dichalcogenide Semiconductor. *Nat. Mater.* **2014**, *13*, 1091−1095.
(39) Liang, L.; Wang, J.; Lin, W.; Sumpter, B. G.; Meunier, V.; Pan, M. Electronic Bandgap and Edge Reconstruction in Phosphorene Materials. *Nano Lett.* **2014**, *14*, 6400−6406.
(40) Lu, C. P.; Li, G.; Mao, J.; Wang, L. M.; Andrei, E. Y. Bandgap and doping effects in MoS2 measured by Scanning Tunneling Microscopy and Spectroscopy. *Nano Lett.* **2014**, *14*, 4628−4633.
(41) Andrei, E. Y.; Li, G.; Du, X. Electronic properties of graphene: a perspective from scanning tunneling microscopy and magneto-transport. *Rep. Prog. Phys.* **2012**, *75*, 056501.
(42) Luican, A.; Li, G.; Andrei, E. Y. Scanning tunneling microscopy and spectroscopy of graphene layers on graphite. *Solid State Commun.* **2009**, *149*, 1151−1156.
(43) Li, G.; Andrei, E. Y. Observation of Landau levels of Dirac fermions in graphite. *Nat. Phys.* **2007**, *3*, 623−627.
(44) Burstein, E.; Lundqvist, S.; *Tunneling Phenomena in Solids*; Plenum Press: New York, 1969.
(45) Zhang, C.; Chen, Y.; Johnson, A.; Li, M.-Y.; Li, L.-J.; Mende, P. C.; Feenstra, R. M.; Shih, C.-K. Probing Critical Point Energies of Transition Metal Dichalcogenides: Surprising Indirect Gap of Single Layer WSe_2. *Nano Lett.* **2015**, *15*, 6494.
(46) Choi, S.; Shaolin, Z.; Yang, W. Layer-Number-Dependent Work Function of MoS_2 Nanoflakes. *J. Korean Phys. Soc.* **2014**, *64*, 1550.
(47) Britnell, L.; Ribeiro, R.; Eckmann, A.; Jalil, R.; Belle, B.; Mishchenko, A.; Kim, Y.-J.; Gorbachev, R.; Georgiou, T.; Morozov, S.; et al. Strong Light−Matter Interactions in Heterostructures of Atomically Thin Films. *Science* **2013**, *340*, 1311.
(48) Shi, H.; Pan, H.; Zhang, Y.-W.; Yakobson, B. I. Quasiparticle band structures and optical properties of strained monolayer MoS2 and WS2. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2013**, *87*, 155304.
(49) Feenstra, R. M. Electrostatic Potential for a Hyperbolic Probe Tip near a Semiconductor. *J. Vac. Sci. Technol., B: Microelectron. Process. Phenom.* **2003**, *21*, 2080.
(50) Feenstra, R. M.; Meyer, G.; Rieder, K.-H. Transport limitations in tunneling spectroscopy of Ge(111)c(2 × 8) surfaces. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2004**, *69*, 081309.
(51) Feenstra, R. M.; Gaan, S.; Meyer, G.; Rieder, K.-H. Low-temperature tunneling spectroscopy of Ge(111)c(2 × 8) surfaces. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2005**, *71*, 125316.
(52) Chernikov, A.; van der Zande, A.; Hill, H. M.; Rigosi, A. F.; Velauthapillai, A.; Hone, J.; Heinz, T. F. Electrical Tuning of Exciton Binding Energies in Monolayer WS_2. *Phys. Rev. Lett.* **2015**, *115*, 126802.
(53) Kosmider, K.; Fernandez-Rossier, J. Electronic Properties of the MoS_2-WS_2 Heterojunction. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2013**, *87*, 075451.
(54) Komsa, H. P.; Krasheninnikov, A. V. Electronic Structures and Optical Properties of Realistic Transition Metal Dichalcogenide Heterostructures from First Principles. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2013**, *88*, 085318.
(55) Zhang, C.; Johnson, A.; Hsu, C. L.; Li, L. J.; Shih, C. K. Direct Imaging of Band Profile in Single Layer MoS2 on Graphite: Quasiparticle Energy Gap, Metallic Edge States, and Edge Band Bending. *Nano Lett.* **2014**, *14*, 2443.
(56) Liu, X.; Balla, I.; Bergeron, H.; Campbell, G. P.; Bedzyk, M. J.; Hersam, M. C. Rotationally Commensurate Growth of MoS_2 on Epitaxial Graphene. *ACS Nano* **2016**, *10*, 1067.
(57) Lin, Y.-C.; Ghosh, R. K.; Addou, R.; Lu, N.; Eichfeld, S. M.; Zhu, H.; Li, M.-Y.; Peng, X.; Kim, M. J.; Li, L.-J.; Wallace, R. M.; Datta, S.; Robinson, J. A. Atomically Thin Resonant Tunnel Diodes Built from Synthetic van der Waals Heterostructures. *Nat. Commun.* **2015**, *6*, 7311.

DOI: 10.1021/acs.nanolett.6b01007
Nano Lett. 2016, 16, 4831−4837

(58) Liu, H.-L.; Shen, C.-C.; Su, S.-H.; Hsu, C.-L.; Li, M.-Y.; Li, L.-J. Optical Properties of Monolayer Transition Metal Dichalcogenides Probed by Spectroscopic Ellipsometry. *Appl. Phys. Lett.* **2014**, *105*, 201905.
(59) Peimyoo, N.; Shang, J.; Cong, C.; Shen, X.; Wu, X.; Yeow, E. K. L.; Yu, T. Nonblinking, Intense Two-Dimensional Light Emitter: Monolayer WS2 Triangles. *ACS Nano* **2013**, *7*, 10985−10994.

DOI: 10.1021/acs.nanolett.6b01007
Nano Lett. 2016, 16, 4831−4837